

Entropy's Edge v9 — Implementation Specification & Mathematical Supplement

This document specifies the full implementation design and theoretical underpinnings of *Entropy's Edge*, a 4X turn-based strategy game grounded in Relativistic Scalar–Vector Plenum (RSVP) cosmology. It integrates thermodynamic field theory, ethics simulation, and game mechanics into one coherent entropic model.

1. Conceptual Foundation

RSVP Concept	Game Mechanic
Φ (scalar potential / capacity)	Local resource richness and habitable potential.
$\mathbf{\Psi}$ (vector flow)	Trade, migration, or energy flow routes; coherence optimization.
S (entropy field)	System disorder or informational smoothness; lower entropy yields higher precision.
Lamphron–Lamphrodyne	Turn phases of expansion vs. consolidation.
Expyrosis	Late-game freeze where players can seed a new cycle.
TARTAN tiling	Recursive plenum grid with memory retention.

2. Turn Structure

Each turn alternates between exploration, expansion, exploitation, extermination, and rebalancing. The Lamphron–Lamphrodyne meta-cycle modulates diffusion and damping coefficients, alternating entropy descent and relaxation.

3. Field Equations (RSVP Dynamics)

The RSVP dynamics governing each tile are given by coupled PDEs:

$$\begin{aligned} \partial_t \Phi &= \kappa \nabla^2 \Phi - \lambda S + \eta \Phi \mathbf{\Psi}(x,t) \\ \partial_t S &= \kappa S \nabla^2 S + \gamma |\nabla \Phi|^2 - \mu S S + \eta S \mathbf{\Psi}(x,t) \\ \partial_t \mathbf{\Psi} &= \kappa \nabla (\nabla \cdot \mathbf{\Psi} - \nabla^2 \mathbf{\Psi}) - \nabla S - \mu \nabla \mathbf{\Psi} \end{aligned}$$

4. Mission Chains and Ethics Dynamics

Missions evolve as Markov processes across an ethics manifold. Each stage transitions according to the potential overlap of Φ – S gradients and ethical field alignment E_{ethic} .

$$P_{\{i \rightarrow j\}} = \sigma(\alpha \mathbf{\Psi} \cdot \nabla \Phi + \beta E_{\text{ethic}} - \gamma E_{\text{torsion}})$$

5. Fleet Combat Cards

Fleets represent mobile condensates following gradient flows. Their attributes derive from the local plenum state: $M = \Phi(1-S)$, $F = ||\mathbf{F}||$, $E = \lambda S$. Combat outcomes draw random cards modifying these multipliers (e.g. Focus Beam, Mirror Burst, Torsion Lock).

6. Scenario Generator

Initial conditions $\Phi_{\mathbf{x}}(x)$, $S_{\mathbf{x}}(x)$ are seeded via correlated noise fields parameterized as Smooth, Chaotic, or Clustered archetypes. AI temperament modifies λ and γ coefficients to simulate aggression or cooperation.

7. Victory Metrics

Entropy Equilibrium (Peace Victory): $G(t) = 1/|\Omega| \int (|\nabla\Phi|^2 + |\nabla S|^2) dx < \epsilon$ for 100 turns. Dominion Victory: control $\geq 70\%$ of map. Rebirth Cycle: simultaneous Expyrosis and Inflaton event.

8. Implementation Layers

Frontend: HTML5/JS with reactive UI and canvas visualization. Simulation kernel: explicit Euler PDE solver in JavaScript. AI/diplomacy: ethical tensor averaging and local field alignment heuristics. Persistence: JSON + browser localStorage. Rendering: Hex/square grid, vector arrows, faction overlays, minimap.

9. Future Implementation Roadmap

1. **Phase 1 – Advanced AI Diplomacy:** Agents learn ethical equilibria via gradient descent on E_i fields. 2. **Phase 2 – Procedural Galaxies:** Fractal generation of initial Φ/S topology via recursive TARTAN morphogenesis. 3. **Phase 3 – Conscious Field Coupling:** Introduce observer-dependent perturbations modeling measurement entropy. 4. **Phase 4 – Cross-Domain Coupling:** Connect RSVP fields to biological cognition simulations (RSVP-AI hybrid). 5. **Phase 5 – Multiplayer Rebirth Campaign:** Synchronize parallel plenum universes across sessions via shared entropy seeds.

10. Philosophical and Physical Rationale

Entropy's Edge interprets the evolution of civilizations as a thermodynamic metaphor: progress equals the smoothing of entropy gradients without total homogenization. Victory corresponds to balance, not domination — a restoration of the plenum's recursive unity.